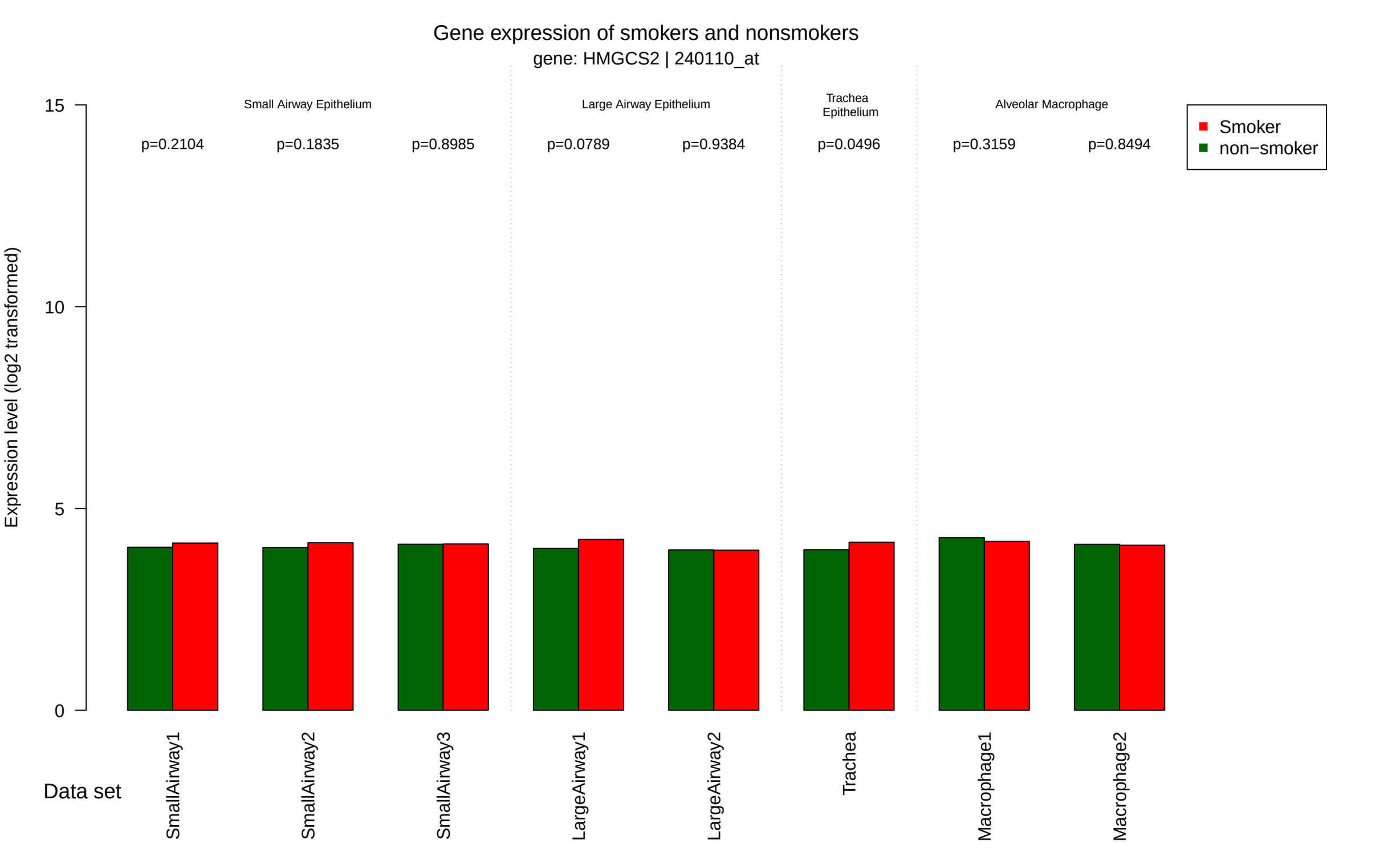
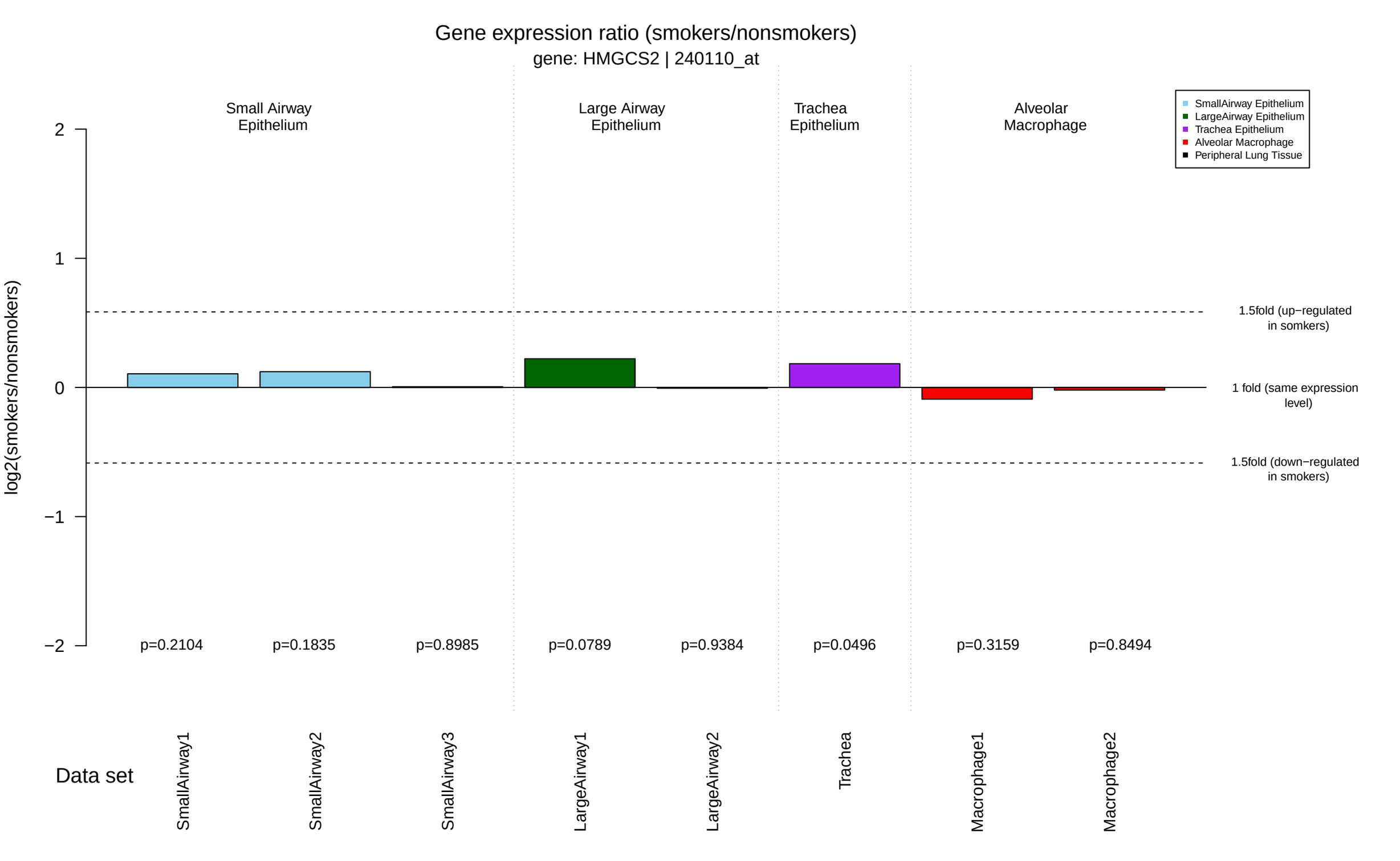
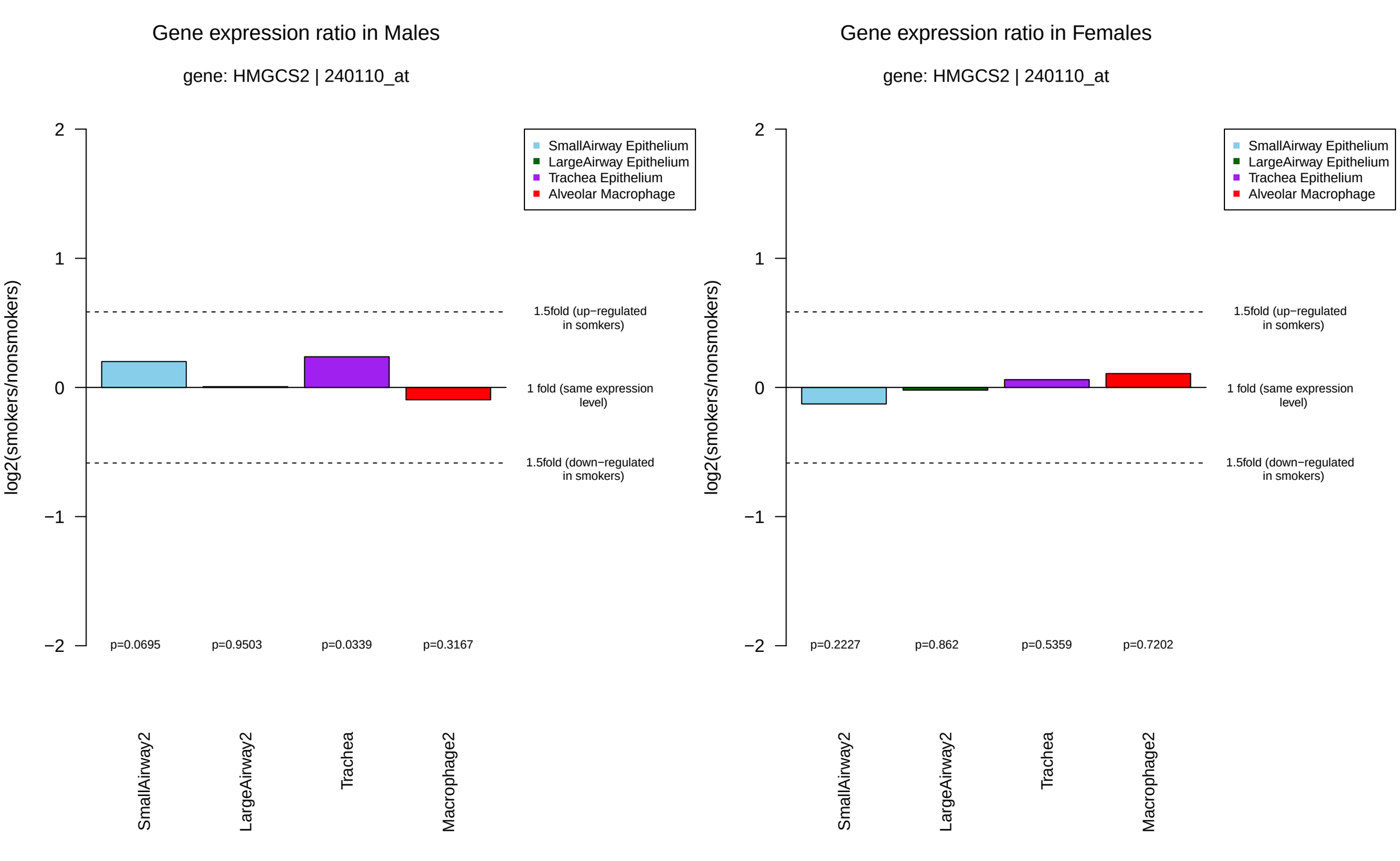
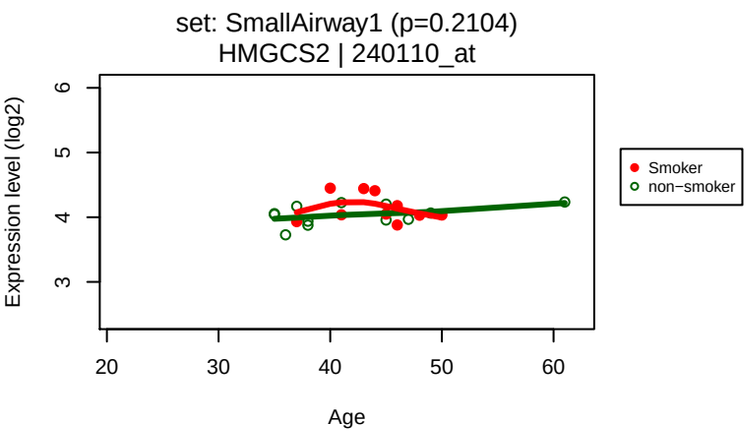
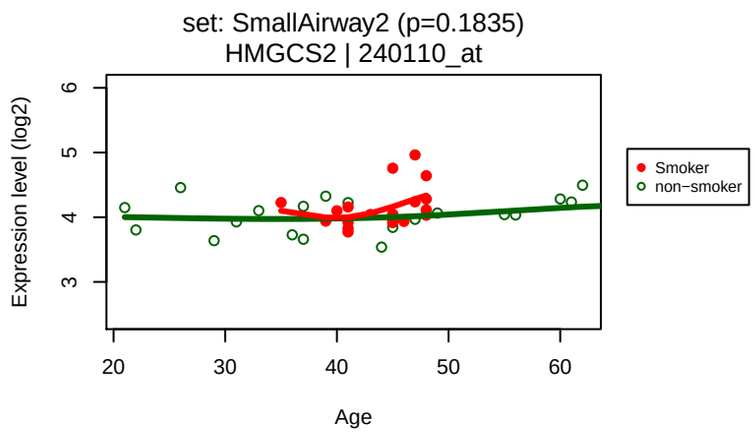
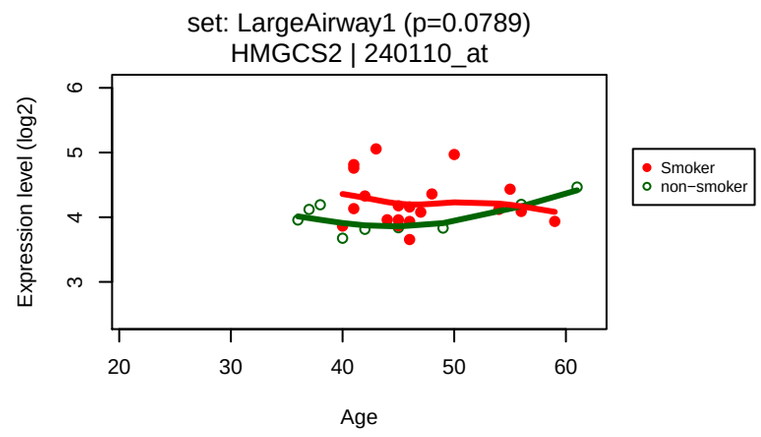

Gene expression of smokers and nonsmokers
gene: HMGCS2 | 240110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2104
p=0.1835
p=0.8985
p=0.0789
p=0.9384
p=0.0496
p=0.3159
p=0.8494
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

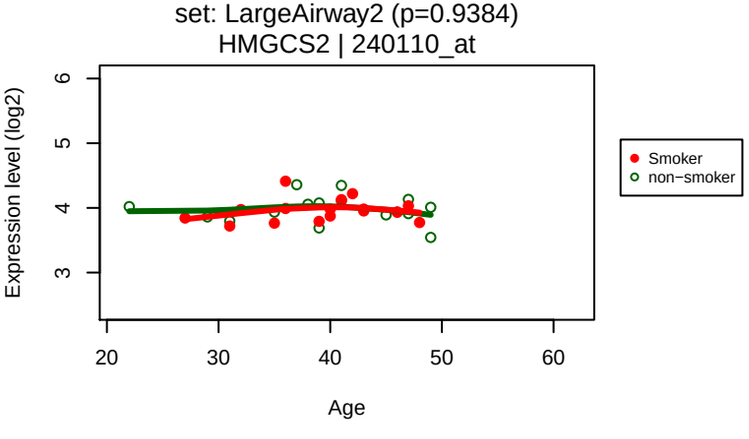

Gene expression ratio (smokers/nonsmokers)
gene: HMGCS2 | 240110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2104
p=0.1835
p=0.8985
p=0.0789
p=0.9384
p=0.0496
p=0.3159
p=0.8494
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

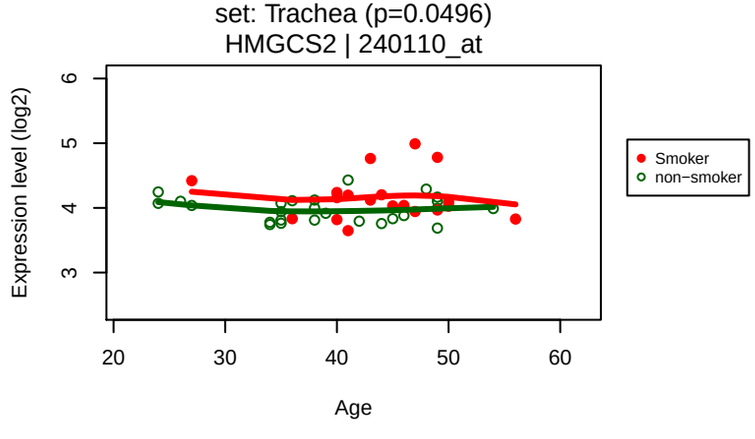

Gene expression ratio in Males
gene: HMGCS2 | 240110_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0695
p=0.9503
p=0.0339
p=0.3167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGCS2 | 240110_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2227
p=0.862
p=0.5359
p=0.7202
SmallAirway2
LargeAirway2
Trachea
Macrophage2

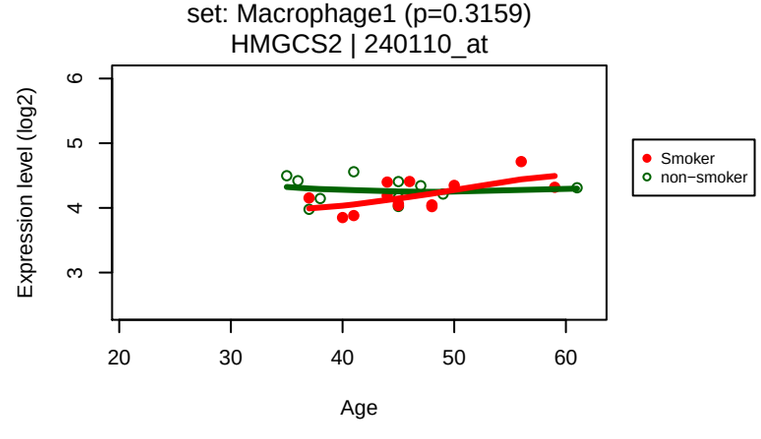

set: SmallAirway1 (p=0.2104)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker

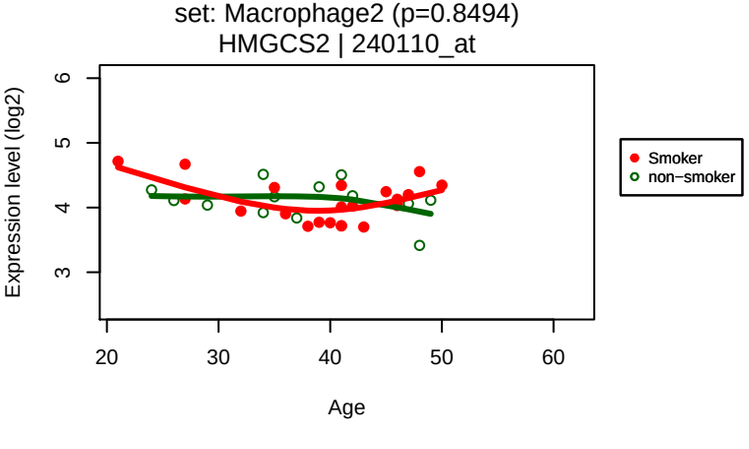

set: SmallAirway2 (p=0.1835)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker

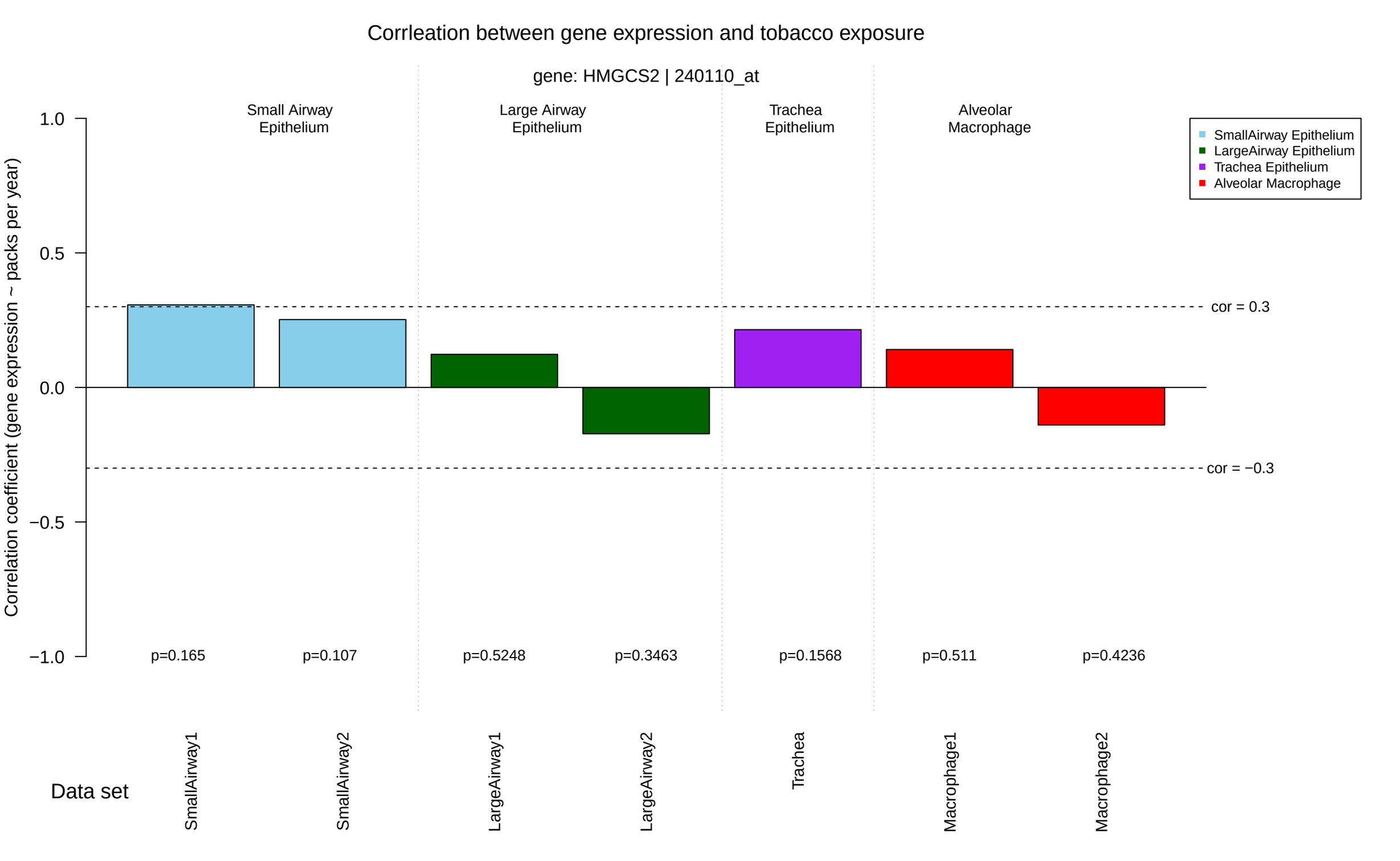

set: LargeAirway1 (p=0.0789)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9384)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0496)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3159)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8494)
HMGCS2 | 240110_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGCS2 | 240110_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.165
p=0.107
p=0.5248
p=0.3463
p=0.1568
p=0.511
p=0.4236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2